

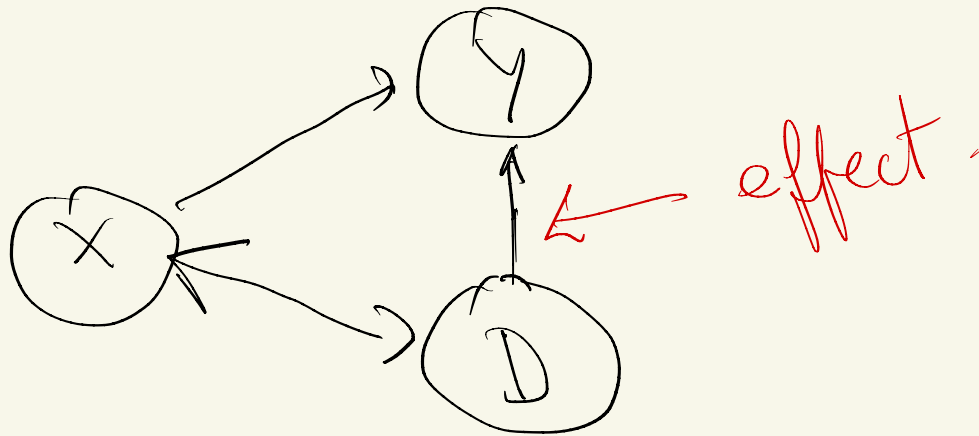
Eco 202 - Fall 2026

Class 05

$Y = \text{outcome}$ (birth weight)

$D = \text{dummy}$ ($= 1$ smoke during
 $= 0$ no-smoke)

$X = \text{confounders}$



$$D_i = \begin{cases} = 0 & \text{no treat} \\ = 1 & \text{treat} \end{cases}$$

$Y_i(1)$ = outcome had i received treatment ($D_i=1$)

$Y_i(0)$ = outcome had i not received treatment ($D_i=0$)

observed outcome →

$$Y_i = Y_i(1) \cdot D_i + Y_i(0) (1 - D_i) = \begin{cases} Y_i(0) & \text{if } D_i=0 \\ Y_i(1) & \text{if } D_i=1 \end{cases}$$